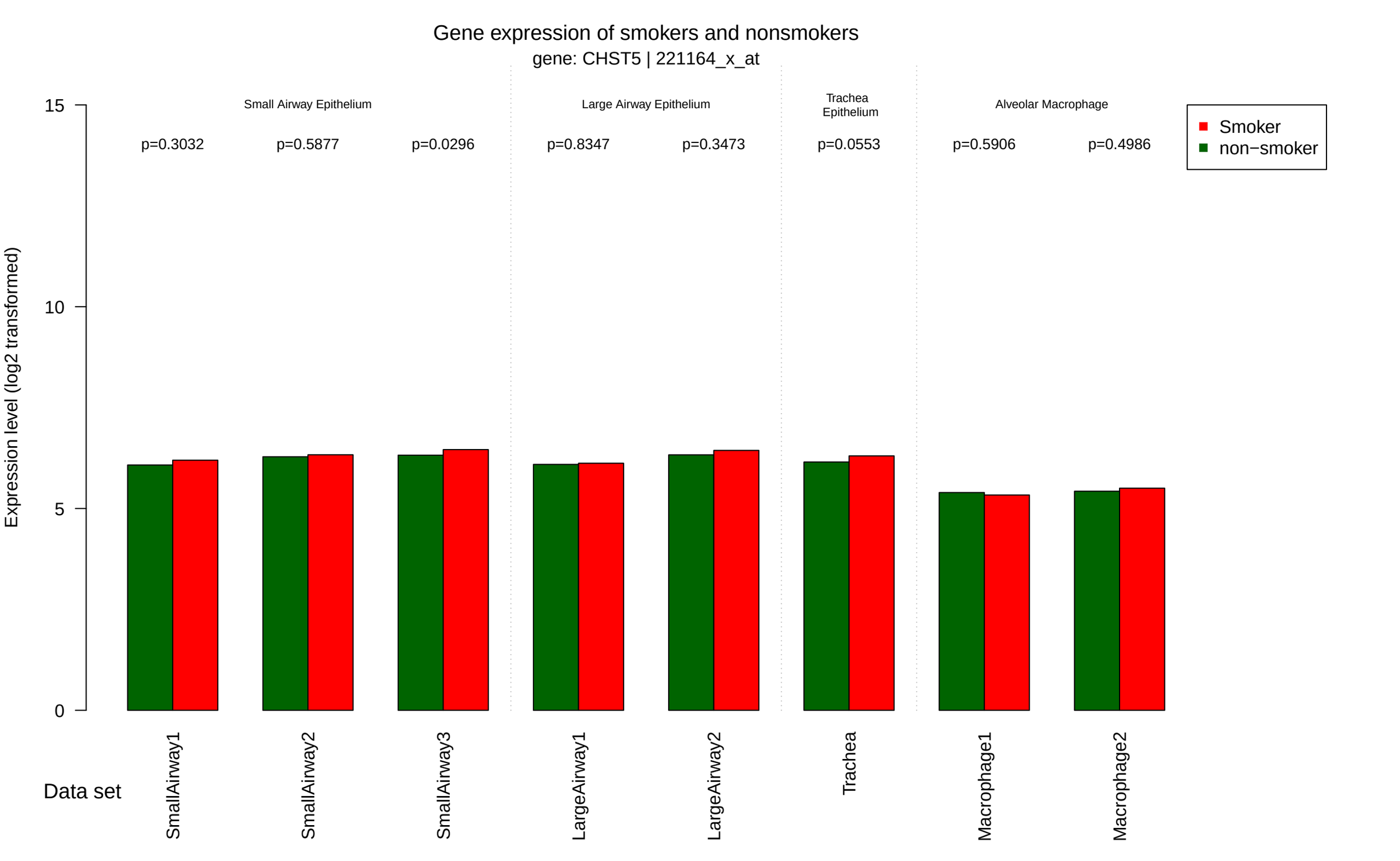
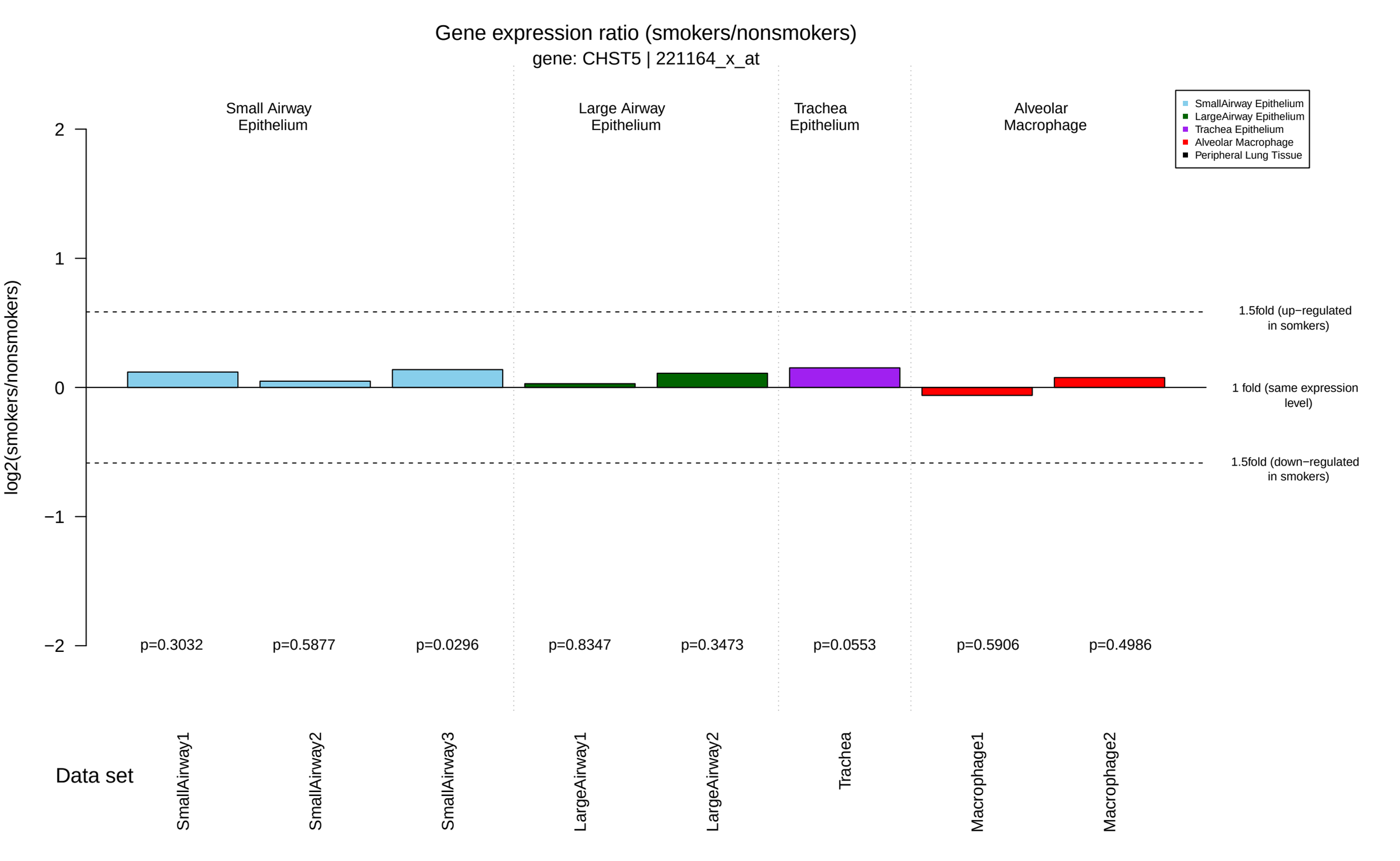
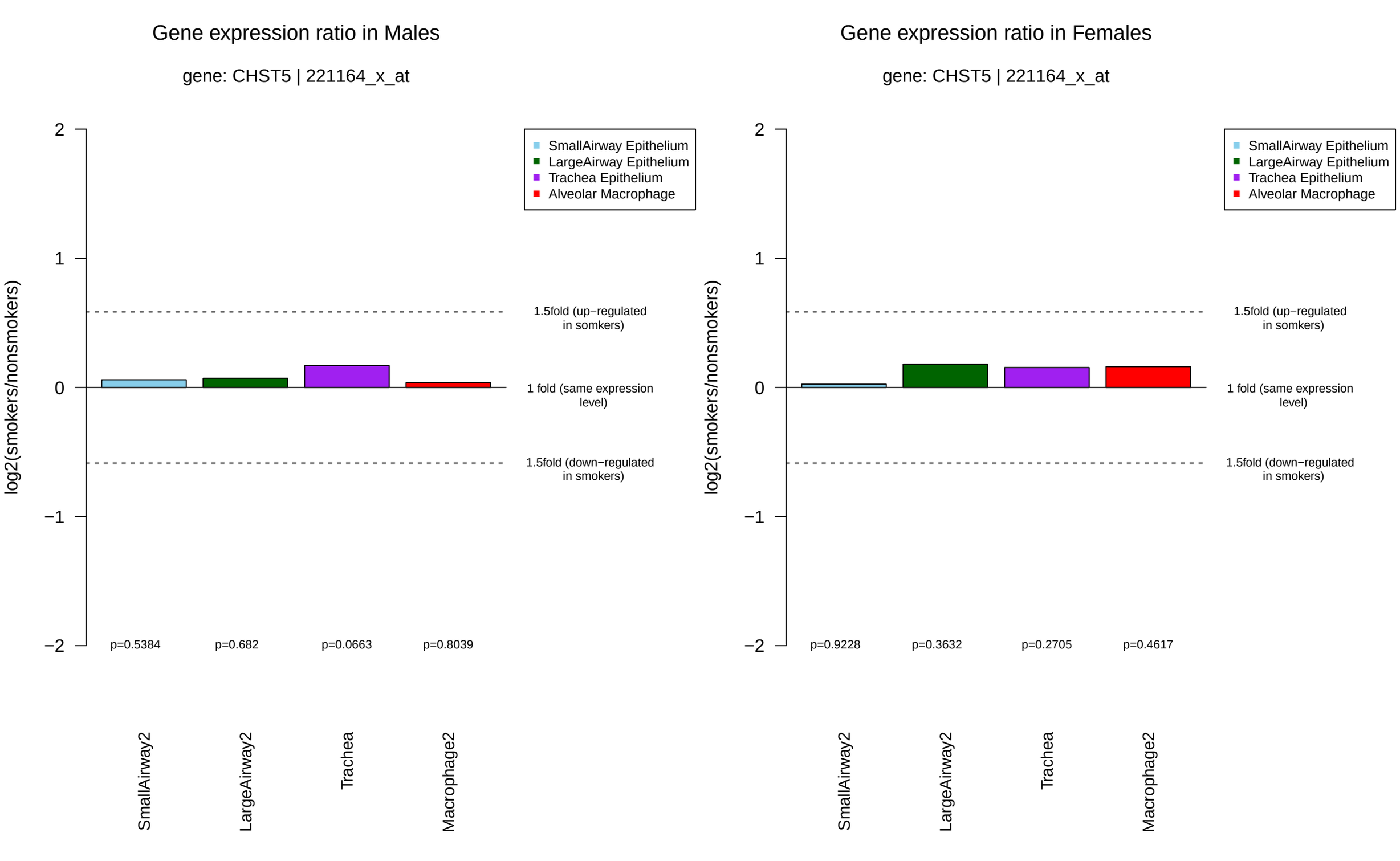
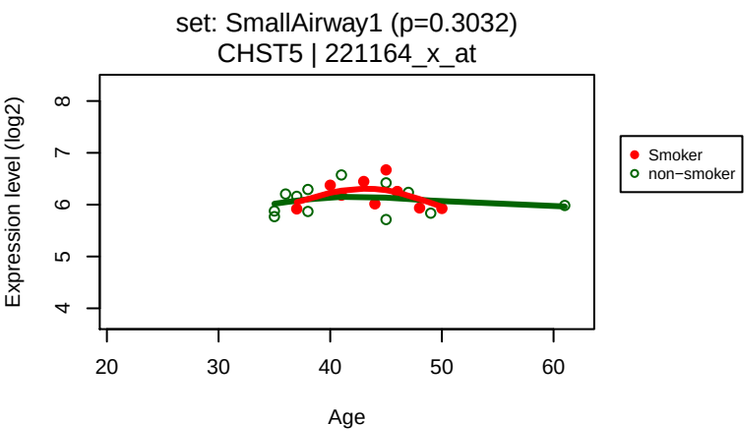
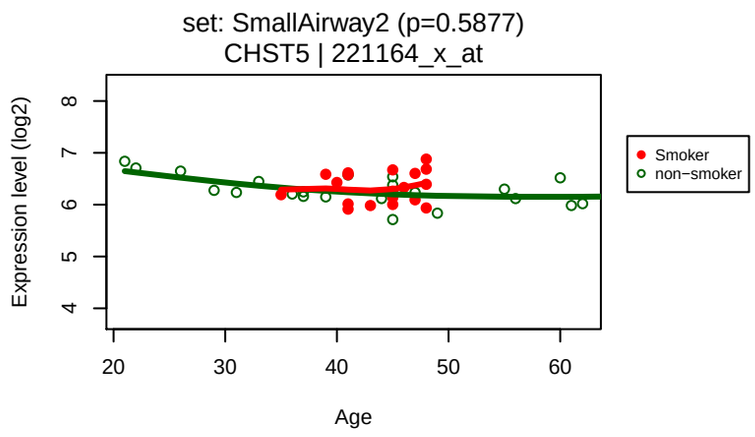
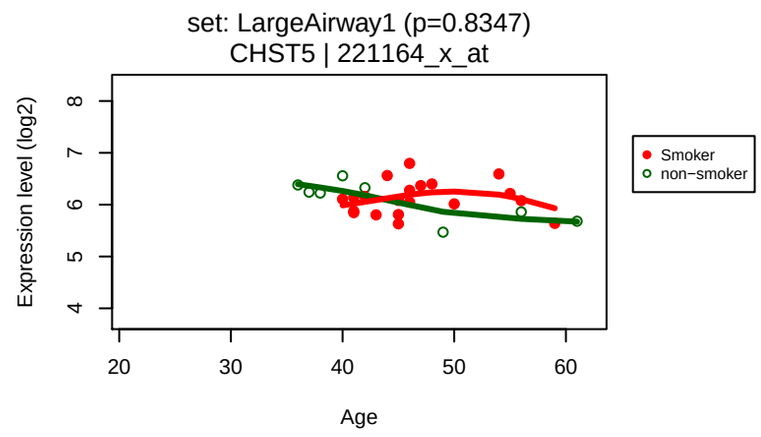

Gene expression of smokers and nonsmokers
gene: CHST5 | 221164_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3032
p=0.5877
p=0.0296
p=0.8347
p=0.3473
p=0.0553
p=0.5906
p=0.4986
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

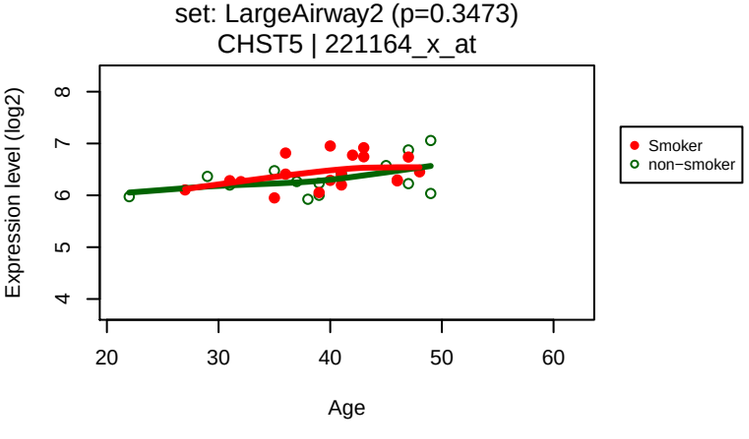

Gene expression ratio (smokers/nonsmokers)
gene: CHST5 | 221164_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3032
p=0.5877
p=0.0296
p=0.8347
p=0.3473
p=0.0553
p=0.5906
p=0.4986
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

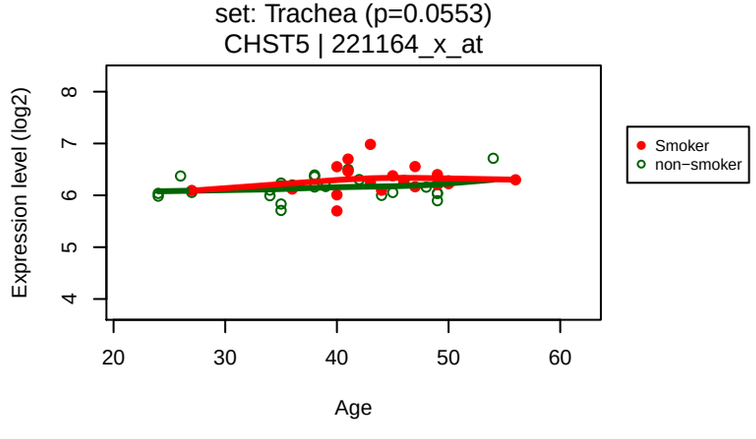

Gene expression ratio in Males
gene: CHST5 | 221164_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5384
p=0.682
p=0.0663
p=0.8039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHST5 | 221164_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9228
p=0.3632
p=0.2705
p=0.4617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

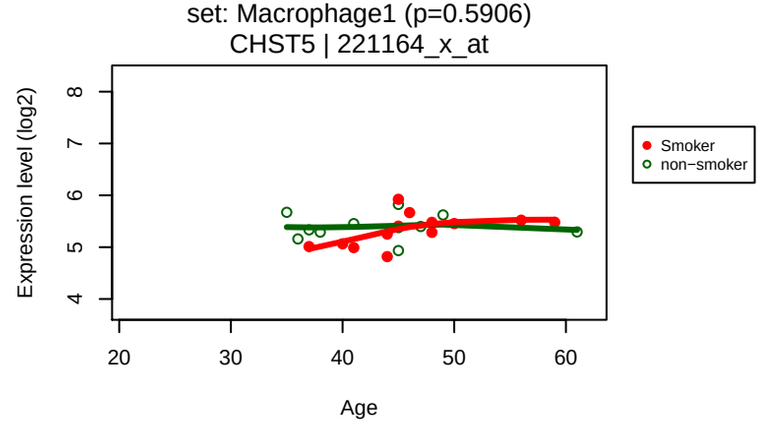

set: SmallAirway1 (p=0.3032)
CHST5 | 221164_x_at
Expression level (log2)
Age
Smoker
non−smoker

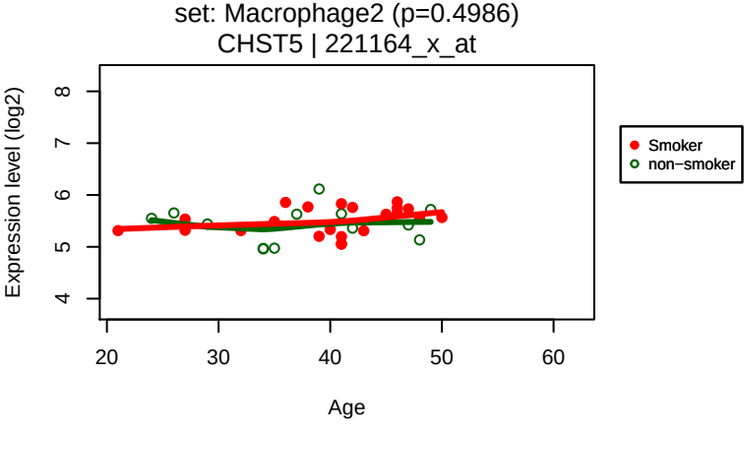

set: SmallAirway2 (p=0.5877)
CHST5 | 221164_x_at
Expression level (log2)
Age
Smoker
non−smoker

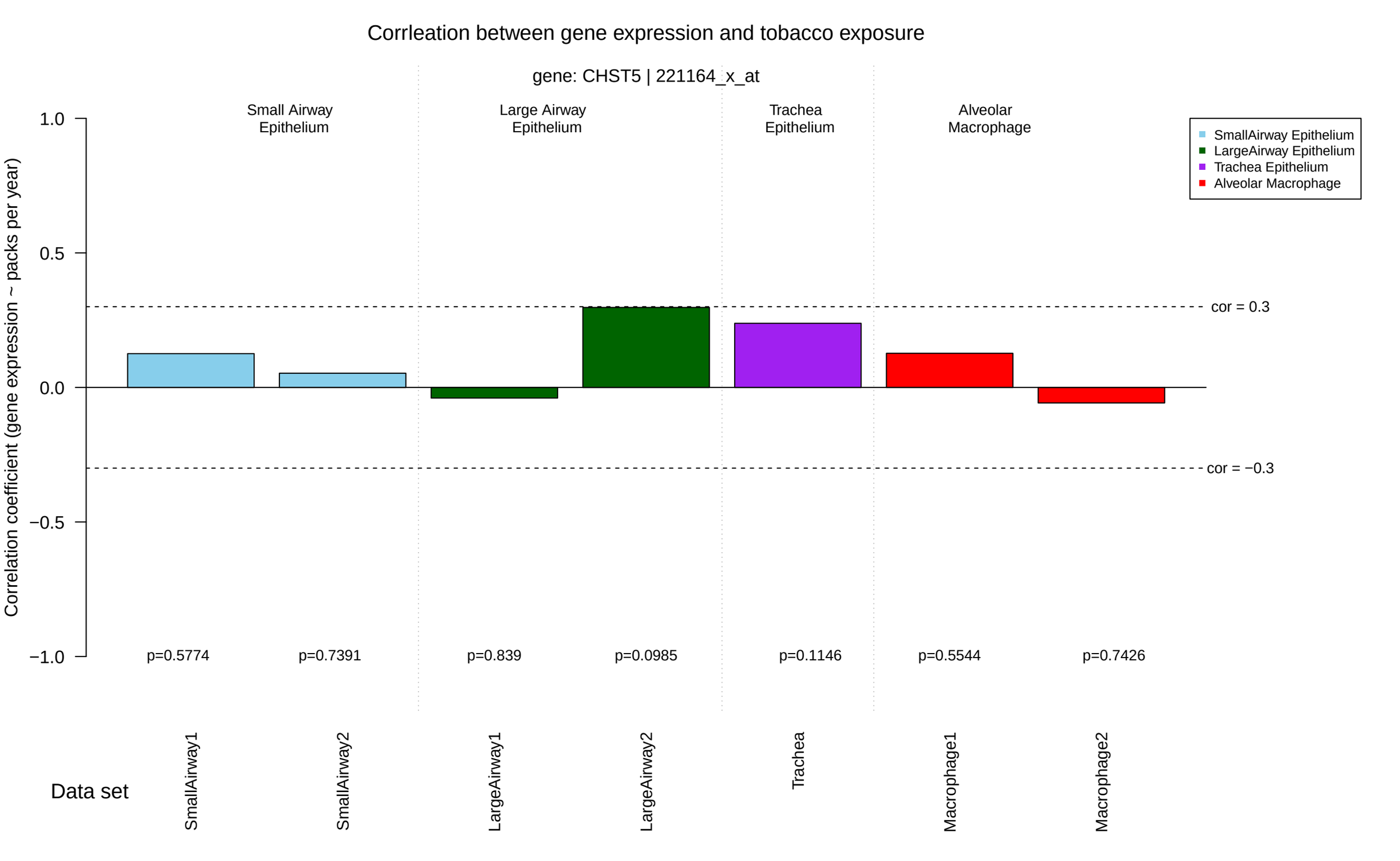

set: LargeAirway1 (p=0.8347)
CHST5 | 221164_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3473)
CHST5 | 221164_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0553)
CHST5 | 221164_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5906)
CHST5 | 221164_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4986)
CHST5 | 221164_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHST5 | 221164_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5774
p=0.7391
p=0.839
p=0.0985
p=0.1146
p=0.5544
p=0.7426
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2